

01 - Burgers' Equation Experiments

AC-PINN Project | Authors: Suyash Vasal Jain, Nishita Raghvendra

PDE: $u_t + u u_x = \nu u_{xx}$, $\nu = 0.01/\pi$

IC: $u(x,0) = -\sin(\pi x)$ | **BC:** $u(-1,t) = u(1,t) = 0$

Domain: $x \in [-1,1]$, $t \in [0,1]$

Architecture: [2, 64, 64, 64, 64, 64, 1] | **Epochs:** 10000

Experiments

1. Vanilla PINN - clean dense data
2. Vanilla PINN - noisy sparse data ($\epsilon=0.1$, $N_{ic}=20$)
3. AC-PINN - clean dense data
4. AC-PINN - noisy sparse data ($\epsilon=0.1$, $N_{ic}=20$)

```
In [1]: import sys, os
sys.path.append('.')
import torch
import numpy as np
import matplotlib.pyplot as plt
from pinn_base import (
    device, NoisyDataGenerator, PINNSolver, ACPINNSolver,
    BurgersFDM, Benchmark, save_metrics, save_history, save_training_plots
)

PDE = 'burgers'
LAYERS = [2, 64, 64, 64, 64, 64, 1]
EPOCHS = 10000
PDE_PARAMS = {'nu': 0.01/np.pi}
RESULTS = '../results/burgers/'
FIGURES = '../figures/burgers/'
os.makedirs(RESULTS, exist_ok=True)
os.makedirs(FIGURES, exist_ok=True)

gen = NoisyDataGenerator(pde=PDE, **PDE_PARAMS)
print(f'Device: {device}')
```

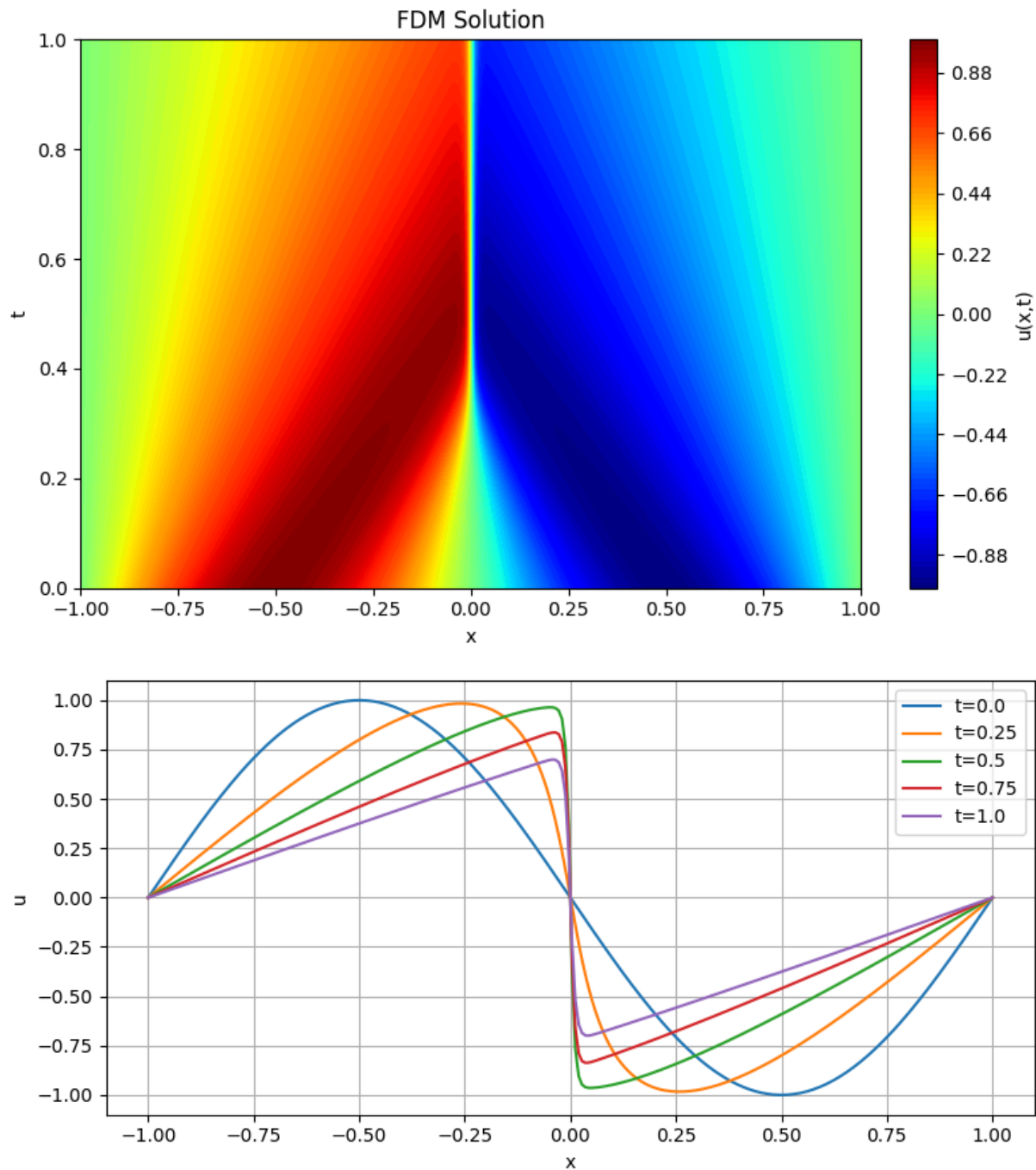
Device: cuda

Section 1 - FDM Ground Truth

```
In [2]: fdm = BurgersFDM(nx=256, nt=2000, nu=0.01/np.pi)
fdm.solve()
fdm.plot_solution()
```

```
fdm.plot_time_slices()
print(f'FDM runtime: {fdm.runtime:.4f}s')
```

BurgersFDM solved in 0.5670s



FDM runtime: 0.5670s

Section 2 - Data Conditions

```
In [3]: # Clean dense
data_clean_dense = gen.generate(N_ic=1000, N_bc=1000, N_f=8000, noise_eps=0.0)
# Noisy sparse
data_noisy_sparse = gen.generate(N_ic=20, N_bc=20, N_f=2000, noise_eps=0.1)

print('Data conditions ready')
```

```
print(f' Clean dense : N_ic={data_clean_dense["x_ic"].shape[0]}')
print(f' Noisy sparse : N_ic={data_noisy_sparse["x_ic"].shape[0]}, eps=0.1')
```

Data conditions ready

Clean dense : N_ic=1000

Noisy sparse : N_ic=20, eps=0.1

Section 3 - Experiment 1: Vanilla PINN, Clean Dense

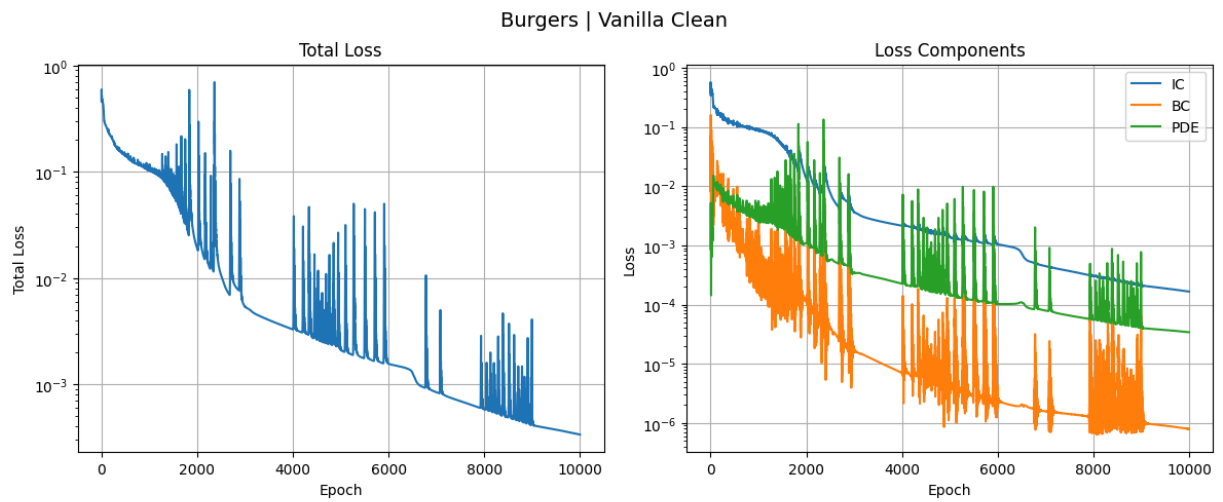
```
In [4]: vanilla_clean = PINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                                   lambda_ic=1.0, lambda_bc=1.0, lambda_pde=5.0)
h_vc = vanilla_clean.fit(data_clean_dense, epochs=EPOCHS, print_every=1000, label='
save_training_plots(h_vc, FIGURES+'vanilla_clean_training.png', 'Burgers | Vanilla
vanilla_clean.plot_loss_history(h_vc)
vanilla_clean.plot_solution(title='Vanilla PINN - Clean Dense')
vanilla_clean.plot_initial_condition_comparison(gen)
save_history(h_vc, RESULTS + 'vanilla_clean_history.npy')
```

D:\PINN\ac-pinn-project\ac-pinn-project\venv\Lib\site-packages\torch\autograd\graph.py:825: UserWarning: Attempting to run cuBLAS, but there was no current CUDA context! Attempting to set the primary context... (Triggered internally at C:\actions-runner_work\pytorch\pytorch\builder\windows\pytorch\aten\src\ATen\cuda\CublasHandlePool.cpp:135.)

return Variable._execution_engine.run_backward(# Calls into the C++ engine to run the backward pass

[Burgers Vanilla Clean Epoch	0/10000]	Total: 0.466202	IC: 0.391538	BC: 0.070330	PDE: 0.000867
[Burgers Vanilla Clean Epoch	1000/10000]	Total: 0.103735	IC: 0.086455	BC: 0.001064	PDE: 0.003243
[Burgers Vanilla Clean Epoch	2000/10000]	Total: 0.018998	IC: 0.013434	BC: 0.000109	PDE: 0.001091
[Burgers Vanilla Clean Epoch	3000/10000]	Total: 0.005401	IC: 0.003638	BC: 0.000016	PDE: 0.000349
[Burgers Vanilla Clean Epoch	4000/10000]	Total: 0.003285	IC: 0.002145	BC: 0.000007	PDE: 0.000227
[Burgers Vanilla Clean Epoch	5000/10000]	Total: 0.002161	IC: 0.001378	BC: 0.000006	PDE: 0.000155
[Burgers Vanilla Clean Epoch	6000/10000]	Total: 0.001562	IC: 0.001047	BC: 0.000003	PDE: 0.000102
[Burgers Vanilla Clean Epoch	7000/10000]	Total: 0.000843	IC: 0.000446	BC: 0.000002	PDE: 0.000079
[Burgers Vanilla Clean Epoch	8000/10000]	Total: 0.000585	IC: 0.000305	BC: 0.000001	PDE: 0.000056
[Burgers Vanilla Clean Epoch	9000/10000]	Total: 0.000469	IC: 0.000205	BC: 0.000004	PDE: 0.000052

Training complete in 152.12s

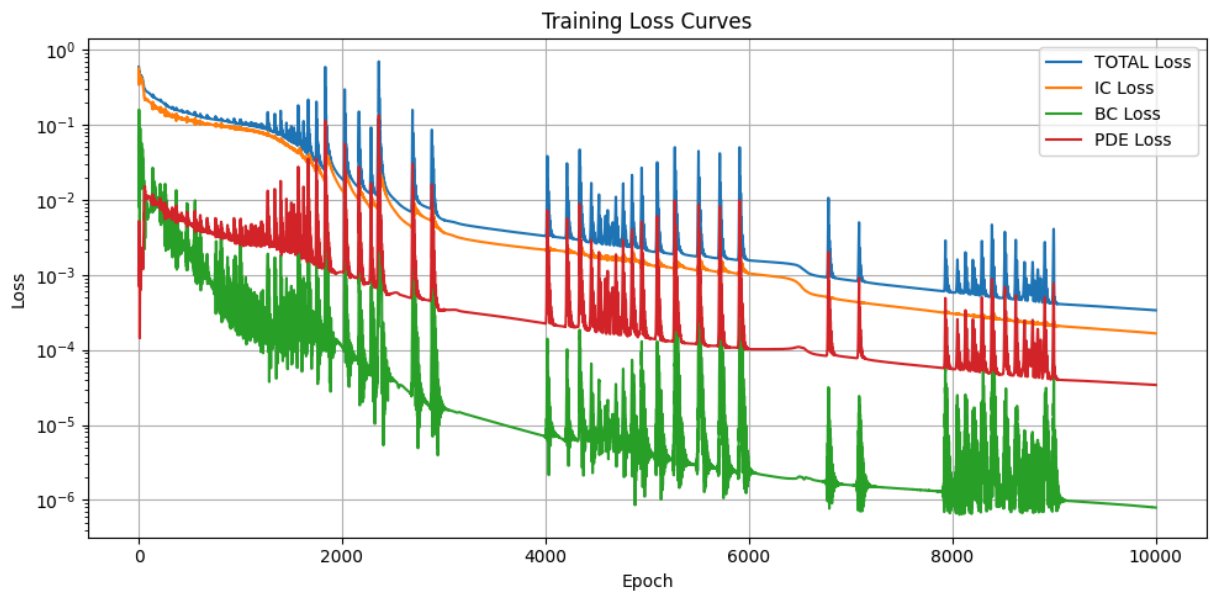


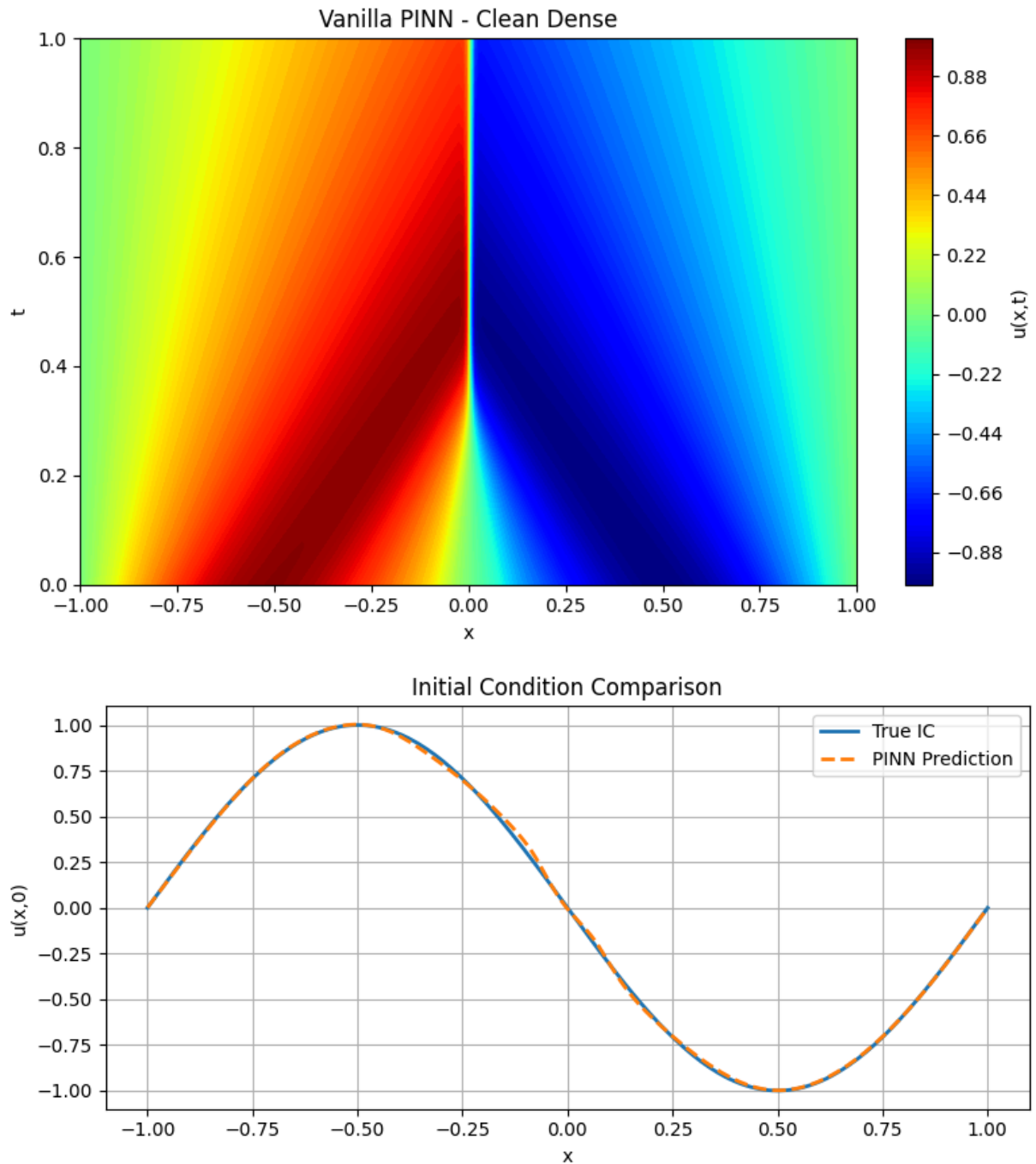
N/A - Vanilla PINN

N/A - Vanilla PINN

Saved: ../figures/burgers/vanilla_clean_training.png

Saved: ../figures/burgers/vanilla_clean_training.csv





Saved: ../results/burgers/vanilla_clean_history.npy

Section 4 - Experiment 2: Vanilla PINN, Noisy Sparse

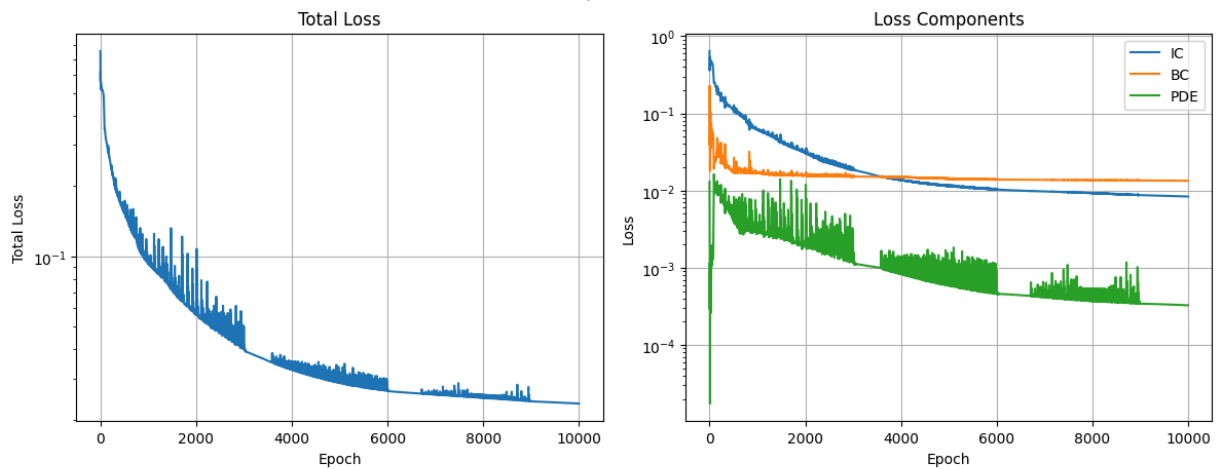
```
In [5]: vanilla_noisy = PINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                                   lambda_ic=1.0, lambda_bc=1.0, lambda_pde=5.0)
h_vn = vanilla_noisy.fit(data_noisy_sparse, epochs=EPOCHS, print_every=1000, label=
save_training_plots(h_vn, FIGURES+'vanilla_noisy_training.png', 'Burgers | Vanilla
vanilla_noisy.plot_loss_history(h_vn)
vanilla_noisy.plot_solution(title='Vanilla PINN - Noisy Sparse')
vanilla_noisy.plot_initial_condition_comparison(gen)
save_history(h_vn, RESULTS + 'vanilla_noisy_history.npy')
```

```

[Burgers | Vanilla Noisy | Epoch    0/10000] Total: 0.755816 | IC: 0.639245 | BC:
0.051520 | PDE: 0.013010
[Burgers | Vanilla Noisy | Epoch  1000/10000] Total: 0.093500 | IC: 0.062155 | BC:
0.016879 | PDE: 0.002893
[Burgers | Vanilla Noisy | Epoch  2000/10000] Total: 0.056005 | IC: 0.030812 | BC:
0.015652 | PDE: 0.001908
[Burgers | Vanilla Noisy | Epoch  3000/10000] Total: 0.045010 | IC: 0.019141 | BC:
0.015512 | PDE: 0.002071
[Burgers | Vanilla Noisy | Epoch  4000/10000] Total: 0.032967 | IC: 0.013674 | BC:
0.014862 | PDE: 0.000886
[Burgers | Vanilla Noisy | Epoch  5000/10000] Total: 0.029428 | IC: 0.011300 | BC:
0.014391 | PDE: 0.000747
[Burgers | Vanilla Noisy | Epoch  6000/10000] Total: 0.026908 | IC: 0.010295 | BC:
0.013903 | PDE: 0.000542
[Burgers | Vanilla Noisy | Epoch  7000/10000] Total: 0.025736 | IC: 0.009755 | BC:
0.013758 | PDE: 0.000445
[Burgers | Vanilla Noisy | Epoch  8000/10000] Total: 0.025565 | IC: 0.009235 | BC:
0.013635 | PDE: 0.000539
[Burgers | Vanilla Noisy | Epoch  9000/10000] Total: 0.023950 | IC: 0.008722 | BC:
0.013514 | PDE: 0.000343
Training complete in 133.18s

```

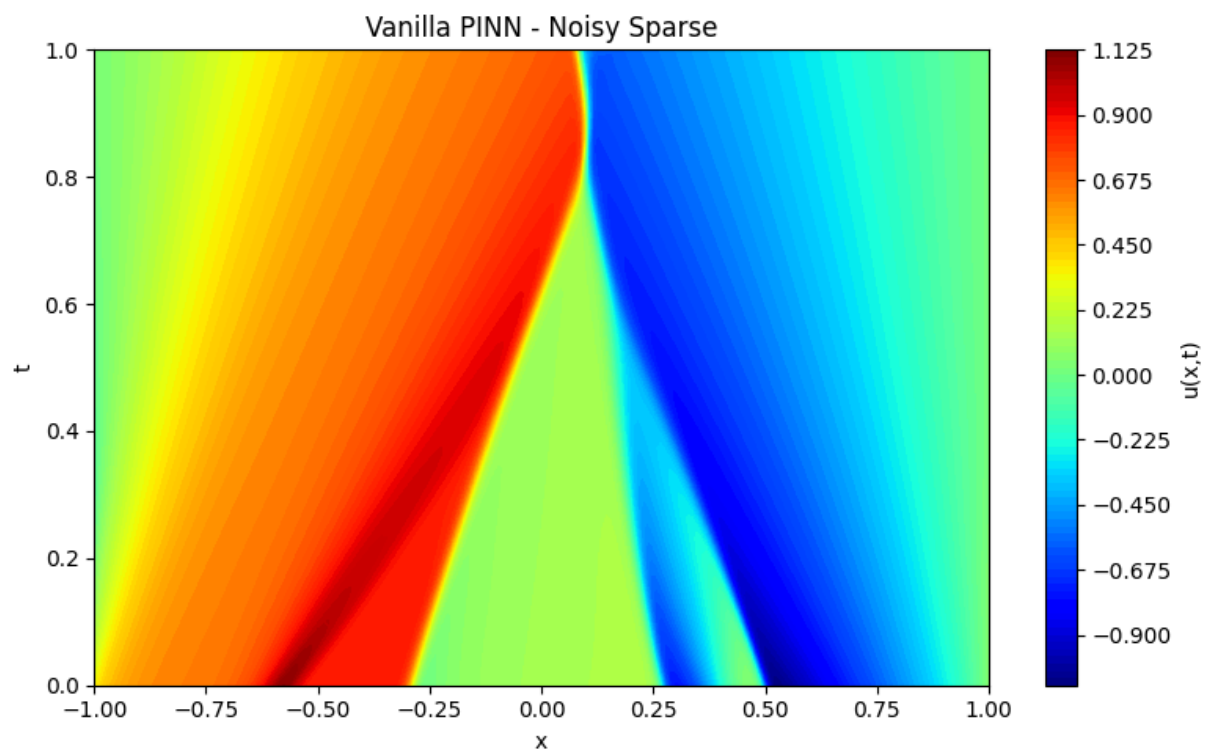
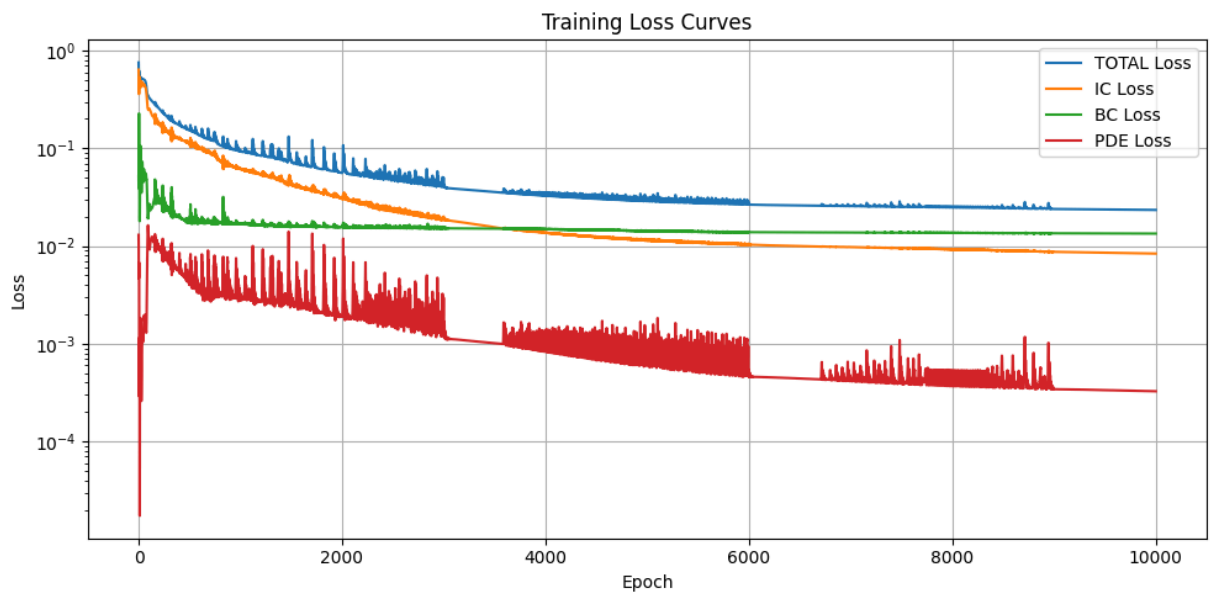
Burgers | Vanilla Noisy

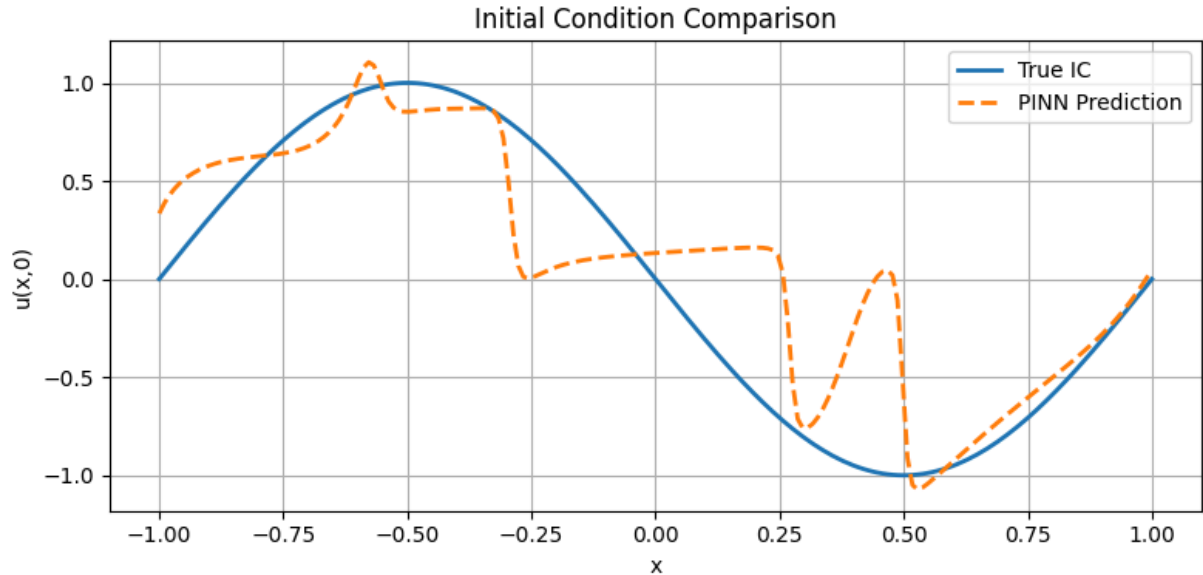


N/A - Vanilla PINN

N/A - Vanilla PINN

Saved: ../figures/burgers/vanilla_noisy_training.png
 Saved: ../figures/burgers/vanilla_noisy_training.csv





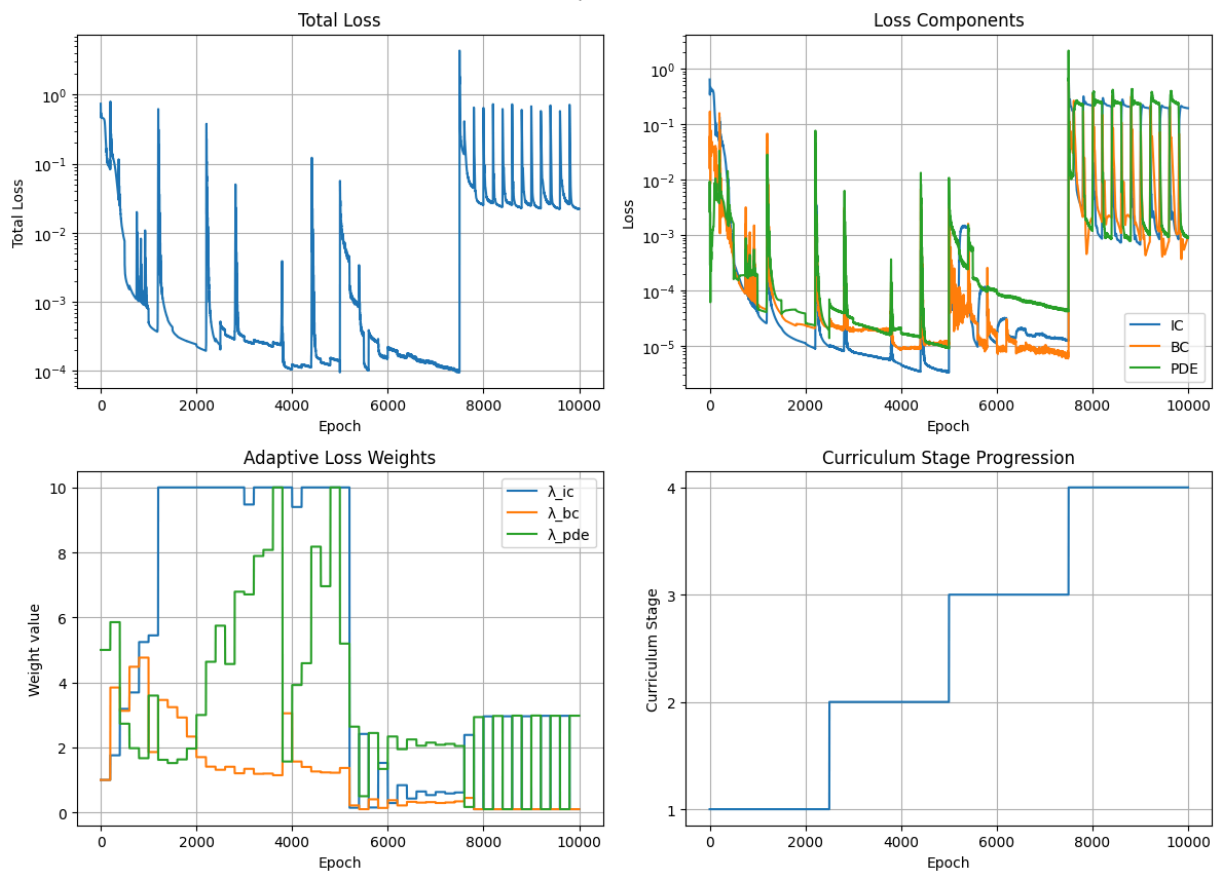
Saved: ../results/burgers/vanilla_noisy_history.npy

Section 5 - Experiment 3: AC-PINN, Clean Dense

```
In [6]: ac_clean = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                                weight_strategy='both', N_pool=20000, resample_every=500)
h_ac = ac_clean.fit(data_clean_dense, epochs=EPOCHS, print_every=1000, label='Burge
save_training_plots(h_ac, FIGURES+'acpinn_clean_training.png', 'Burgers | AC-PINN C
ac_clean.plot_loss_history(h_ac)
ac_clean.plot_weight_history(h_ac)
ac_clean.plot_solution(title='AC-PINN - Clean Dense')
ac_clean.plot_initial_condition_comparison(gen)
save_history(h_ac, RESULTS + 'ac_clean_history.npy')
```

```
[Burgers | AC-PINN Clean | Epoch    0/10000] Stage 1/4 | Total: 0.739858 | IC: 0.64
0984 | BC: 0.058730 | PDE: 0.008029 |  $\lambda=(1.00,1.00,5.00)$ 
[Burgers | AC-PINN Clean | Epoch 1000/10000] Stage 1/4 | Total: 0.000515 | IC: 0.00
0042 | BC: 0.000056 | PDE: 0.000051 |  $\lambda=(5.45,1.86,3.59)$ 
[Burgers | AC-PINN Clean | Epoch 2000/10000] Stage 1/4 | Total: 0.000233 | IC: 0.00
0011 | BC: 0.000024 | PDE: 0.000027 |  $\lambda=(10.00,1.70,3.00)$ 
[Burgers | AC-PINN Clean | Epoch 3000/10000] Stage 2/4 | Total: 0.000279 | IC: 0.00
0008 | BC: 0.000018 | PDE: 0.000027 |  $\lambda=(9.47,1.34,6.71)$ 
[Burgers | AC-PINN Clean | Epoch 4000/10000] Stage 2/4 | Total: 0.000124 | IC: 0.00
0005 | BC: 0.000009 | PDE: 0.000017 |  $\lambda=(9.40,1.57,3.92)$ 
[Burgers | AC-PINN Clean | Epoch 5000/10000] Stage 2/4 | Total: 0.000095 | IC: 0.00
0003 | BC: 0.000011 | PDE: 0.000009 |  $\lambda=(10.00,1.37,5.19)$ 
[Burgers | AC-PINN Clean | Epoch 6000/10000] Stage 3/4 | Total: 0.000221 | IC: 0.00
0011 | BC: 0.000015 | PDE: 0.000091 |  $\lambda=(0.29,0.38,2.34)$ 
[Burgers | AC-PINN Clean | Epoch 7000/10000] Stage 3/4 | Total: 0.000122 | IC: 0.00
0016 | BC: 0.000007 | PDE: 0.000053 |  $\lambda=(0.63,0.29,2.08)$ 
[Burgers | AC-PINN Clean | Epoch 8000/10000] Stage 4/4 | Total: 0.643336 | IC: 0.21
7572 | BC: 0.002072 | PDE: 0.001212 |  $\lambda=(2.96,0.10,0.10)$ 
[Burgers | AC-PINN Clean | Epoch 9000/10000] Stage 4/4 | Total: 0.678464 | IC: 0.00
0670 | BC: 0.000802 | PDE: 0.227568 |  $\lambda=(0.10,0.10,2.98)$ 
AC-PINN training complete in 174.53s
```

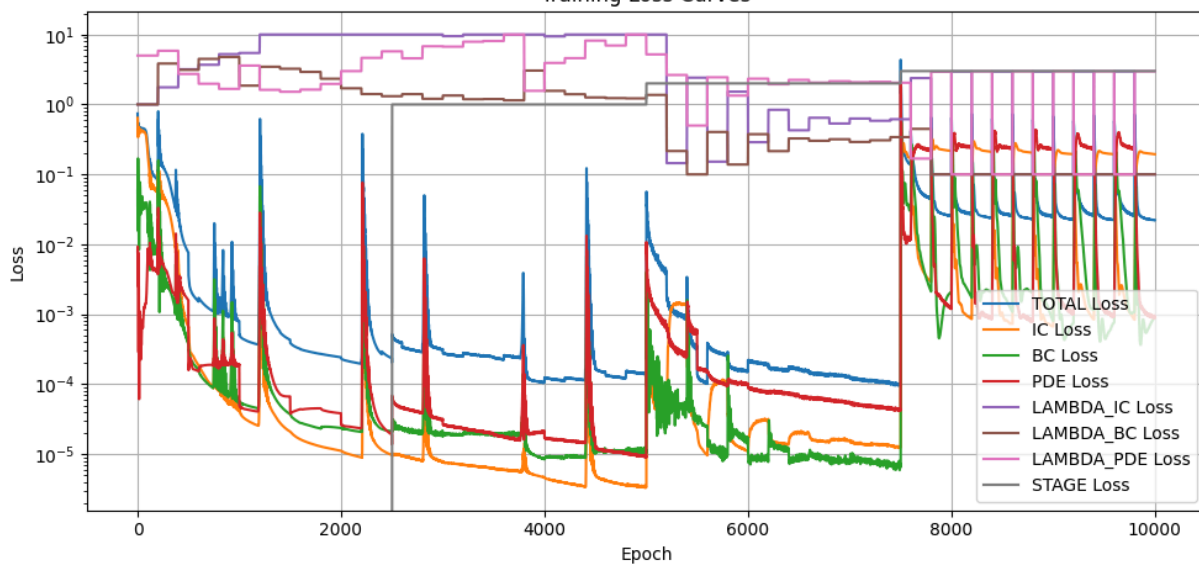

Burgers | AC-PINN Clean

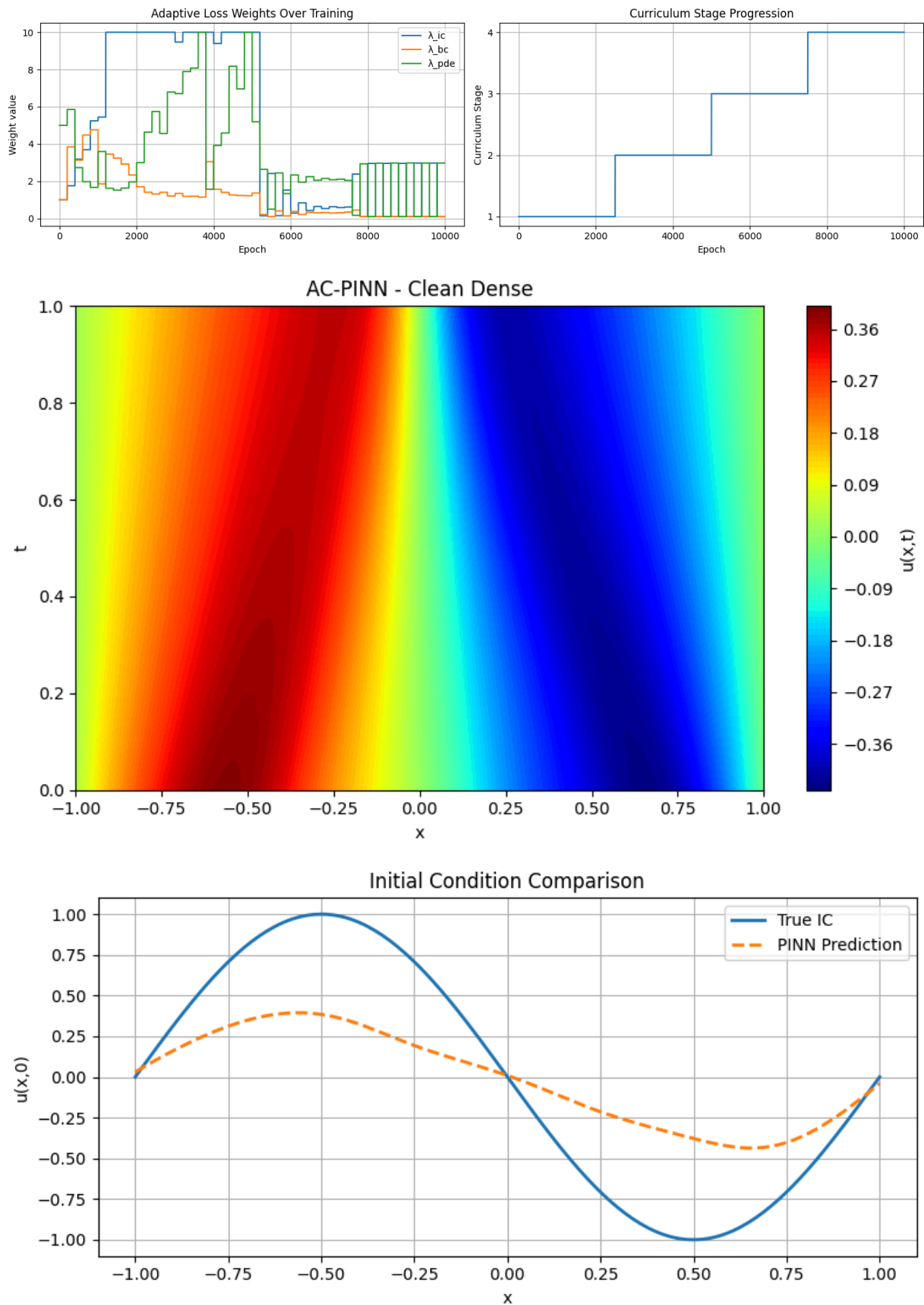


Saved: ../figures/burgers/acpinn_clean_training.png

Saved: ../figures/burgers/acpinn_clean_training.csv

Training Loss Curves





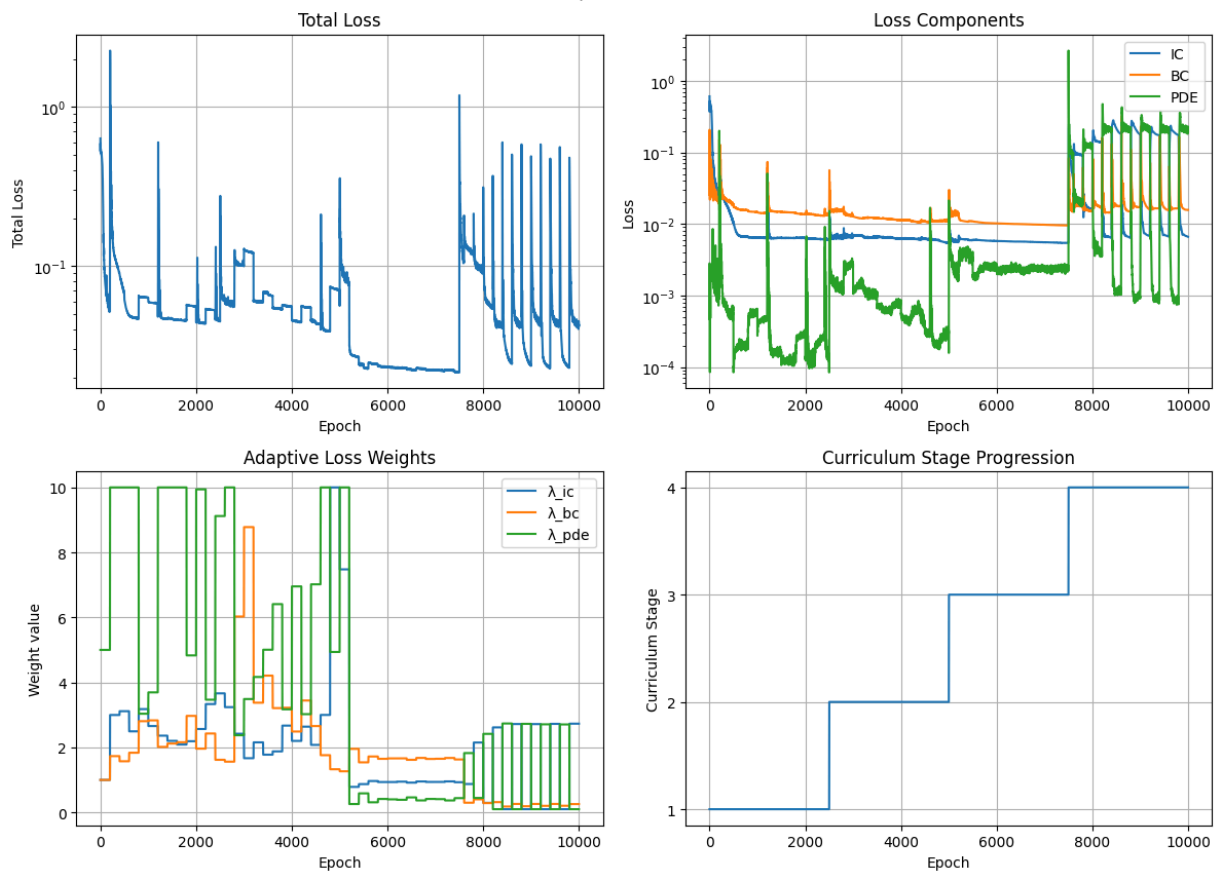
Saved: ../results/burgers/ac_clean_history.npy

Section 6 - Experiment 4: AC-PINN, Noisy Sparse

```
In [7]: ac_noisy = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS,
                                weight_strategy='both', N_pool=20000, resample_every=500)
h_an = ac_noisy.fit(data_noisy_sparse, epochs=EPOCHS, print_every=1000, label='Burgers')
save_training_plots(h_an, FIGURES+'acpinn_noisy_training.png', 'Burgers | AC-PINN Noisy')
ac_noisy.plot_loss_history(h_an)
ac_noisy.plot_weight_history(h_an)
ac_noisy.plot_solution(title='AC-PINN - Noisy Sparse')
ac_noisy.plot_initial_condition_comparison(gen)
save_history(h_an, RESULTS + 'ac_noisy_history.npy')
```

```
[Burgers | AC-PINN Noisy | Epoch    0/10000] Stage 1/4 | Total: 0.633855 | IC: 0.60
5894 | BC: 0.025604 | PDE: 0.000471 |  $\lambda=(1.00,1.00,5.00)$ 
[Burgers | AC-PINN Noisy | Epoch 1000/10000] Stage 1/4 | Total: 0.059711 | IC: 0.00
6398 | BC: 0.014709 | PDE: 0.000282 |  $\lambda=(2.66,2.83,3.69)$ 
[Burgers | AC-PINN Noisy | Epoch 2000/10000] Stage 1/4 | Total: 0.044318 | IC: 0.00
6327 | BC: 0.013670 | PDE: 0.000127 |  $\lambda=(2.57,1.96,9.94)$ 
[Burgers | AC-PINN Noisy | Epoch 3000/10000] Stage 2/4 | Total: 0.128296 | IC: 0.00
6686 | BC: 0.012915 | PDE: 0.001082 |  $\lambda=(1.67,8.78,3.49)$ 
[Burgers | AC-PINN Noisy | Epoch 4000/10000] Stage 2/4 | Total: 0.045917 | IC: 0.00
6245 | BC: 0.011339 | PDE: 0.000568 |  $\lambda=(2.20,2.49,6.95)$ 
[Burgers | AC-PINN Noisy | Epoch 5000/10000] Stage 2/4 | Total: 0.056250 | IC: 0.00
5377 | BC: 0.011423 | PDE: 0.000158 |  $\lambda=(7.48,1.27,10.00)$ 
[Burgers | AC-PINN Noisy | Epoch 6000/10000] Stage 3/4 | Total: 0.023308 | IC: 0.00
5737 | BC: 0.010197 | PDE: 0.002471 |  $\lambda=(0.94,1.66,0.40)$ 
[Burgers | AC-PINN Noisy | Epoch 7000/10000] Stage 3/4 | Total: 0.022263 | IC: 0.00
5536 | BC: 0.009735 | PDE: 0.002416 |  $\lambda=(0.94,1.65,0.41)$ 
[Burgers | AC-PINN Noisy | Epoch 8000/10000] Stage 4/4 | Total: 0.312497 | IC: 0.01
5333 | BC: 0.014940 | PDE: 0.125620 |  $\lambda=(0.30,0.29,2.42)$ 
[Burgers | AC-PINN Noisy | Epoch 9000/10000] Stage 4/4 | Total: 0.489628 | IC: 0.17
8409 | BC: 0.017143 | PDE: 0.001326 |  $\lambda=(2.72,0.26,0.10)$ 
AC-PINN training complete in 145.70s
```

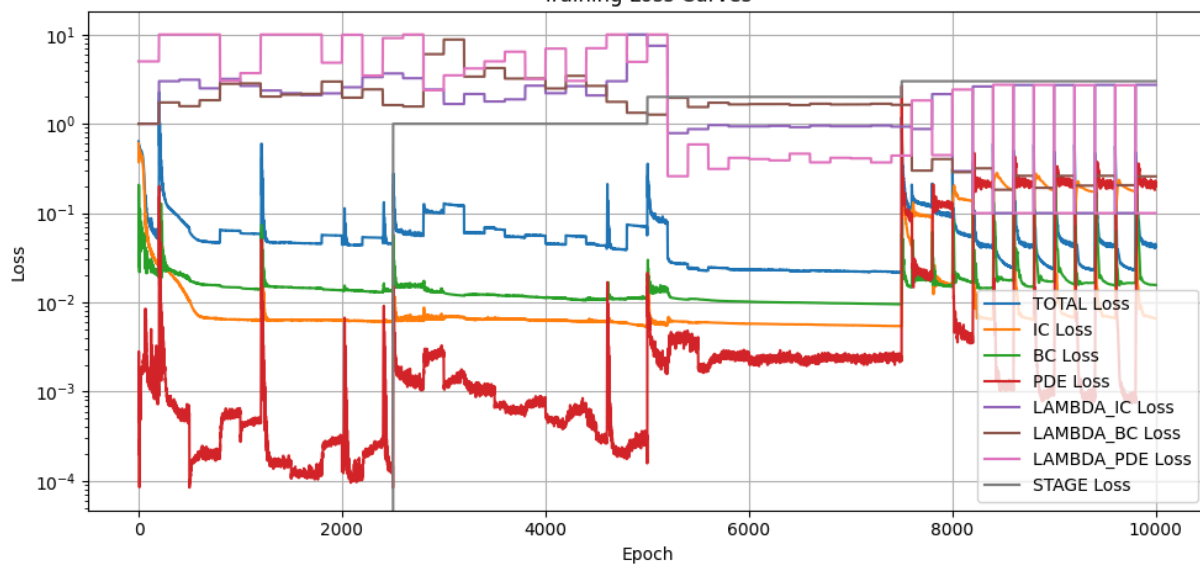
Burgers | AC-PINN Noisy

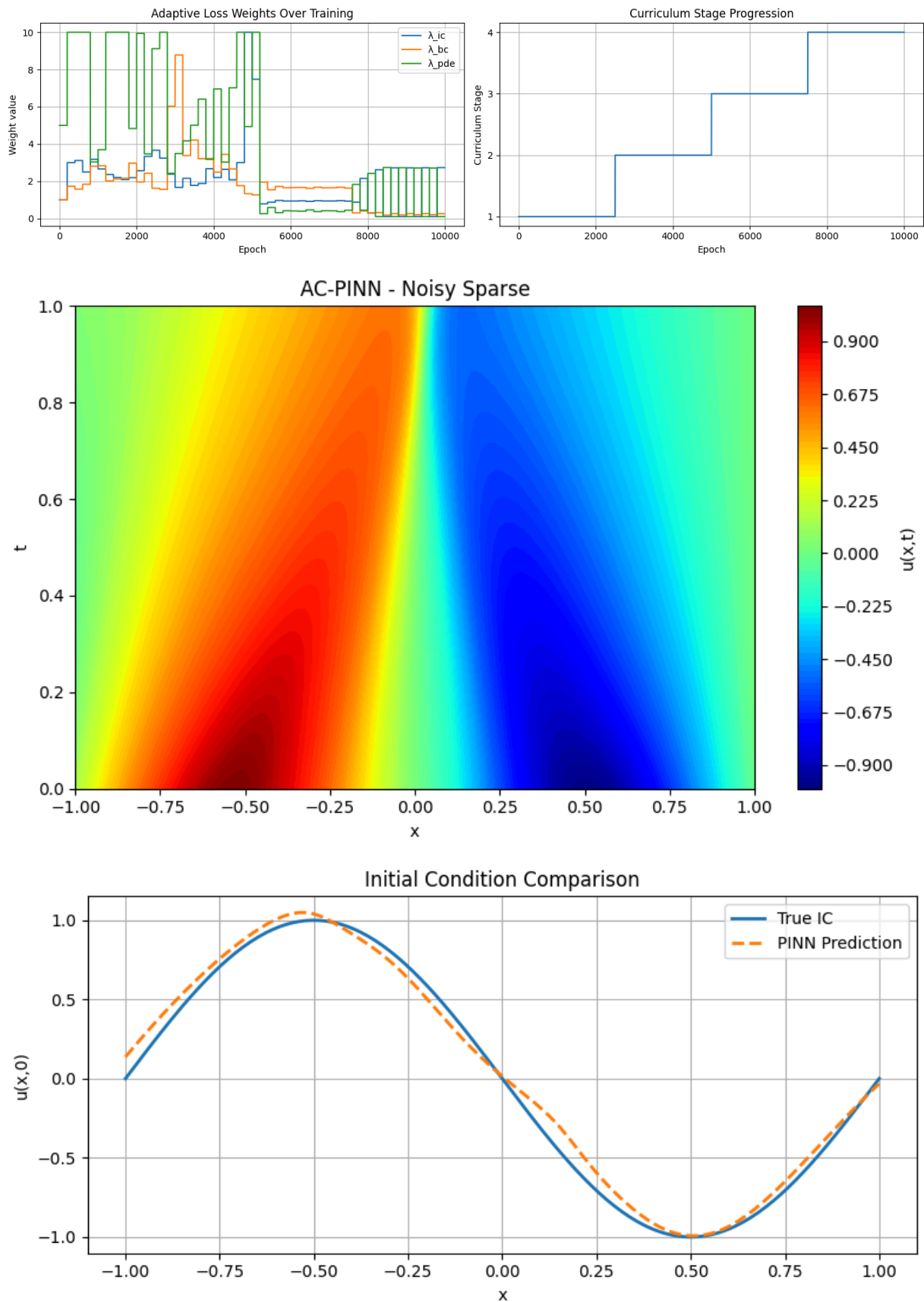


Saved: ../figures/burgers/acpinn_noisy_training.png

Saved: ../figures/burgers/acpinn_noisy_training.csv

Training Loss Curves





Saved: ../results/burgers/ac_noisy_history.npy

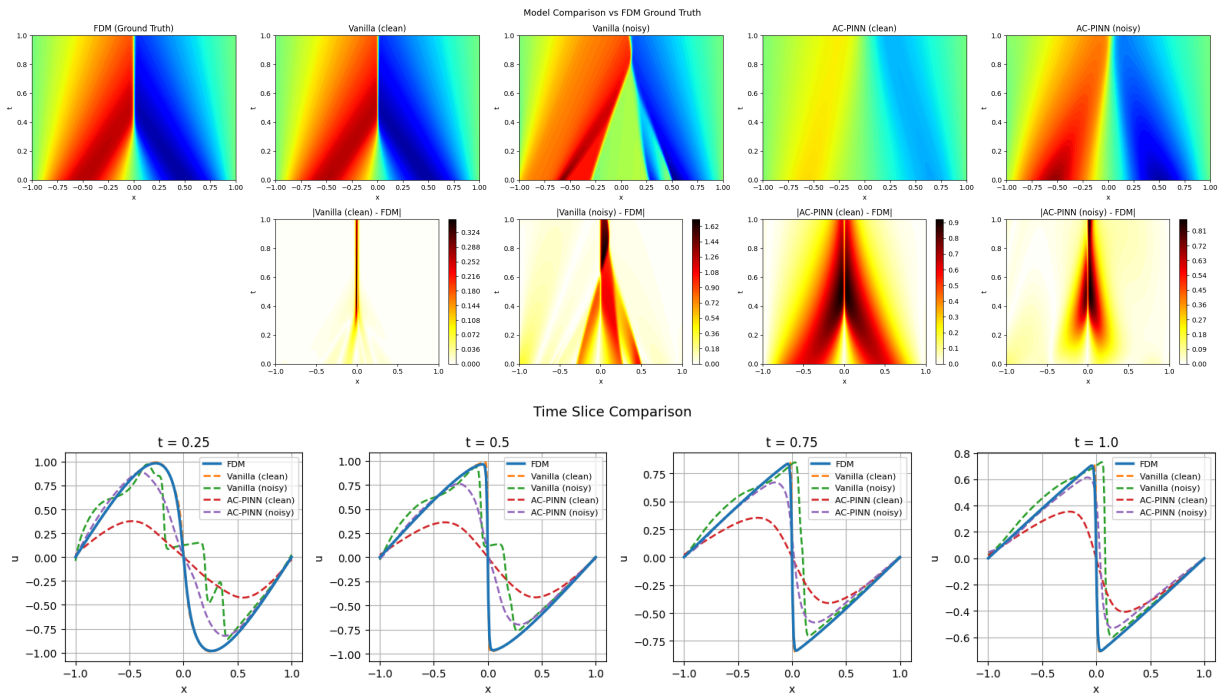
Section 7 - Benchmark vs FDM

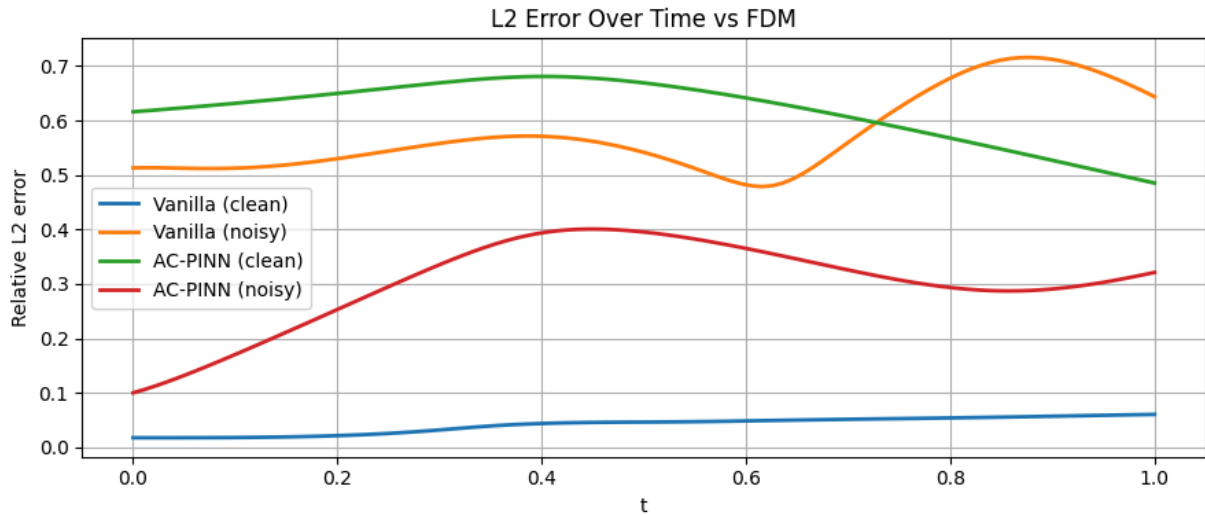
```
In [8]: bench = Benchmark(fdm, nx=200, nt=100)
bench.add('Vanilla (clean)', vanilla_clean)
bench.add('Vanilla (noisy)', vanilla_noisy)
bench.add('AC-PINN (clean)', ac_clean)
bench.add('AC-PINN (noisy)', ac_noisy)
bench.run()

metrics = bench.compare_metrics()
bench.plot_comparison(save_path=FIGURES+'comparison.png')
bench.plot_time_slices(save_path=FIGURES+'time_slices.png')
bench.plot_error_over_time(save_path=FIGURES+'error_over_time.png')

save_metrics(metrics, RESULTS+'benchmark_metrics.npy')
print('Burgers experiments complete. Results saved.')
```

Model	Rel L2	Max Err	MAE	RMSE
Vanilla (clean)	0.040107	0.354410	0.006437	0.024392
Vanilla (noisy)	0.560293	1.683388	0.171400	0.340750
AC-PINN (clean)	0.634461	0.911624	0.290111	0.385857
AC-PINN (noisy)	0.308886	0.875428	0.103675	0.187853





Saved: ../results/burgers/benchmark_metrics.npy
 Burgers experiments complete. Results saved.

Section 8 - Noise Level Study ($\epsilon = 0.05, 0.1, 0.2$)

AC-PINN vs Vanilla under increasing noise.

```
In [9]: noise_results = {}
for eps in [0.05, 0.1, 0.2]:
    print(f'\n--- Noise level  $\epsilon$ ={eps} ---')
    d = gen.generate(N_ic=50, N_bc=50, N_f=3000, noise_eps=eps)

    v = PINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS)
    v.fit(d, epochs=5000, print_every=2500)

    a = ACPINNSolver(pde=PDE, layers=LAYERS, pde_params=PDE_PARAMS, weight_strategy='L2')
    a.fit(d, epochs=5000, print_every=2500)

    b = Benchmark(fdm).add(f'Vanilla  $\epsilon$ ={eps}', v).add(f'AC-PINN  $\epsilon$ ={eps}', a)
    b.run()
    noise_results[eps] = b.compare_metrics()

save_metrics(noise_results, RESULTS+'noise_study_metrics.npy')
print('Noise study complete.')
```

--- Noise level $\epsilon=0.05$ ---

[Epoch 0/5000] Total: 1.063132 | IC: 0.482046 | BC: 0.084301 | PDE: 0.099357

[Epoch 2500/5000] Total: 0.156997 | IC: 0.122314 | BC: 0.007110 | PDE: 0.005515

Training complete in 65.57s

[Epoch 0/5000] Stage 1/4 | Total: 0.417310 | IC: 0.398027 | BC: 0.011612 | PDE: 0.001534 | $\lambda=(1.00, 1.00, 5.00)$

[Epoch 2500/5000] Stage 2/4 | Total: 0.018985 | IC: 0.001621 | BC: 0.004889 | PDE: 0.000131 | $\lambda=(2.92, 2.83, 3.30)$

AC-PINN training complete in 72.23s

Model	Rel L2	Max Err	MAE	RMSE
Vanilla $\epsilon=0.05$	0.265506	1.287047	0.083955	0.161471
AC-PINN $\epsilon=0.05$	0.669988	0.953823	0.314913	0.407463

--- Noise level $\epsilon=0.1$ ---

[Epoch 0/5000] Total: 0.651831 | IC: 0.366401 | BC: 0.157921 | PDE: 0.025502

[Epoch 2500/5000] Total: 0.095362 | IC: 0.058053 | BC: 0.029286 | PDE: 0.001605

Training complete in 66.28s

[Epoch 0/5000] Stage 1/4 | Total: 0.622663 | IC: 0.294049 | BC: 0.326246 | PDE: 0.000474 | $\lambda=(1.00,1.00,5.00)$

[Epoch 2500/5000] Stage 2/4 | Total: 0.178481 | IC: 0.011442 | BC: 0.026912 | PDE: 0.000587 | $\lambda=(10.00,2.34,1.88)$

AC-PINN training complete in 72.04s

Model	Rel L2	Max Err	MAE	RMSE
Vanilla $\epsilon=0.1$	0.195247	0.890570	0.068452	0.118742
AC-PINN $\epsilon=0.1$	0.361080	1.006070	0.123122	0.219596

--- Noise level $\epsilon=0.2$ ---

[Epoch 0/5000] Total: 0.915887 | IC: 0.599887 | BC: 0.139421 | PDE: 0.035316

[Epoch 2500/5000] Total: 0.169403 | IC: 0.070652 | BC: 0.085165 | PDE: 0.002717

Training complete in 65.42s

[Epoch 0/5000] Stage 1/4 | Total: 0.707765 | IC: 0.542642 | BC: 0.136354 | PDE: 0.005754 | $\lambda=(1.00,1.00,5.00)$

[Epoch 2500/5000] Stage 2/4 | Total: 0.314173 | IC: 0.037827 | BC: 0.071384 | PDE: 0.000113 | $\lambda=(3.84,2.36,3.16)$

AC-PINN training complete in 75.63s

Model	Rel L2	Max Err	MAE	RMSE
Vanilla $\epsilon=0.2$	0.344088	1.726001	0.080905	0.209262
AC-PINN $\epsilon=0.2$	0.332351	0.980405	0.121976	0.202124

Saved: ../results/burgers/noise_study_metrics.npy

Noise study complete.